

ACHINTYA RAI

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EDUCATION

University of California, San Diego
Bachelor of Science in Computer Engineering

Sept. 2024 – June 2028
GPA: 3.9/4.0

WORK EXPERIENCE

Adobe
Software Engineering Intern

San Jose, CA
June 2026 – Sept. 2026

- Profiled **60.6M+** payment events across large datasets in Databricks, surfacing **86,387** duplicate records and a **47%** null rate to establish the ground-truth source for anomaly monitoring.
- Designed a time-series anomaly detector on a machine learning model for **30+** countries.
- Automated an MLOps pipeline for weekly retraining and hourly machine learning inference.

Decklar (formerly Roambee)
Software Engineering Intern

Sunnyvale, CA
June 2024 – Aug. 2024

- Developed real-time shipment monitoring modules for **Amazon, Coca-Cola, and Samsung**, optimizing Python and NumPy pipelines for a **15%** capacity boost across **1M+** weekly events.
- Applied advanced data structures and algorithms to analytics, boosting reporting speed **20%**.
- Enhanced workflow efficiency **15%** through Panels-based UI/UX refinements and user testing.

RESEARCH

Graph Theory & Neural Architecture
Undergraduate Research Fellow (NSF REU)

San Diego, CA
March 2026 – Present

- Engineered a Python pipeline to analyze **LLM** embedding manifolds (Llama-2, DeepSeek-V2), compressing **500GB** checkpoints to **13GB** localized manifolds via weight-matrix extraction.
- Optimized preprocessing for a **30x** runtime cut, enabling large-scale analysis on **8GB RAM**.
- Applied Louvain clustering to raise network density from **0.18%** to **13.25%** in Llama-2-70B.

Applications of Multi-Agent Systems
Undergraduate Research Fellow (UC Scholars)

San Diego, CA
June 2025 – Aug. 2025

Multi-agent AI Chess Tutor | *Python, MCP, Django, React, SQLite*

- Architected **multi-agent** workflows cutting hallucinations **80%** with an **83% faster** runtime.
- Engineered custom **MCP** servers standardizing agentic tool access across the framework.
- Developed **7 REST API endpoints** in Django and Python, validated by **15+** chess coaches.

PROJECTS

Watchdog | *Python, Gemini API, Browser-Use, Asyncio, SQLite* 

April 2026

- Engineered an **agentic** browser safeguard using the **browser-use SDK** and **Gemini Vision**.
- Developed a concurrent **Asyncio** CLI for real-time URL blacklisting and **Notion API** syncing.

NanoGPT | *Python, PyTorch, Transformers, Neural Networks* | 

Mar. 2026

- Trained a character-level transformer with **100k+** parameters in **PyTorch** from scratch.
- Engineered scalable dataloaders streaming tokenized datasets at **100%** hardware utilization.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, TypeScript, JavaScript, SQL (Postgres/SQLite), Bash

ML & AI: PyTorch, TensorFlow, Machine Learning, LLMs, Multi-Agent Systems, MLOps, Data Structures & Algorithms

Tools & Cloud: React, Django, FastAPI, Flask, GitHub Copilot, Claude Code, MCP, AWS, GCP, Docker